SAFE INDOOR AIR FOR OREGON SCHOOLS



MCDANIEL HIGH SCHOOL

AIR QUALITY REPORT

Lessons and opportunities for ventilation, air filtration, and clean indoor air across Portland Public Schools

District Airflow Measurements

AMERESCO

Airflow Testing Report

Prepared for

Portland Public Schools

August 2021

Neudorfer Engineers. Inc.
Consulting Engineers Seattle Portland
www.NeudorferEngineers.com

AIRFLOW SURVEY REPORT

Project: Portland Public Schools Airflow Testing
Location: McDaniel HS; 2735 NE 82nd Ave., Portland, OR 97220
Filter Status: Upgraded

Room	Equipment Info		Room Dimensions					Airflow Measurements			Calculated ACH					Notes
	Served By	Equipment Type	Room Length	Room Width	Room Area	Room Height	Room Volume	Total CFM Supply	OA CFM Supply	OA %	Air Changes per Hour (supply)	Air Changes per Hour (OA)	# of Portable Filters	Total Effective Air Changes per Hour (ACH_e) with Portable Filter	Total Effective Air Changes per Hour (ACH_e) without Portable Filter	
Level 0																
Office 031D	FC-E-08	FCU	Note #1	Note #1	93	10.0	930	65	12	18%	4.2	0.8	1	64.4	3.1	
Office 031E	FC-E-08	FCU	Note #1	Note #1	98	10.0	980	65	12	18%	4.0	0.7	1	61.1	2.9	
Conf Rm 029	FC-E-06	FCU	Note #1	Note #1	236	8.8	2,084	245	76	31%	7.1	2.2	1	32.6	5.3	
Rm 027	FC-E-06	FCU	Note #1	Note #1	206	8.8	1,819	225	70	31%	7.4	2.3	1	36.9	5.5	
Rm 015	FC-E-02	FCU	Note #1	Note #1	646	10.0	6,460	370	303	82%	3.4	2.8	1	11.5	2.7	
Rm 013A	FC-E-01	FCU	Note #1	Note #1	979	10.0	9,790	950	460	48%	5.8	2.8	1	10.2	4.4	
Rm 013B	FC-E-03	FCU	Note #1	Note #1	978	10.0	9,780	925	450	49%	5.7	2.8	1	10.1	4.3	
Level 1																
Health 161	FC-A-03	FCU	Note #1	Note #1	153	10.0	1,530	255	26	10%	4.8	0.5	1	21.3	3.5	
Reception 161K	-	-	Note #1	Note #1	167	10.0	1,670	-	-	-						Supply air is provided by Health 161.
Office 161A	FC-A-03	FCU	Note #1	Note #1	120	10.0	1,200	255	26	10%	12.8	1.3	1	56.8	9.3	
Office 161B	FC-A-03	FCU	Note #1	Note #1	161	10.0	1,610	250	25	10%	9.3	0.9	1	42.2	6.8	
Office 161C	FC-A-03	FCU	Note #1	Note #1	121	10.0	1,210	230	23	10%	11.4	1.1	1	55.4	8.3	
Conf Rm 161D	FC-A-01	FCU	Note #1	Note #1	167	10.0	1,670	495	50	10%	17.8	1.8	1	47.1	12.9	
Lab 161E	FC-A-02	FCU	Note #1	Note #1	152	10.0	1,520	460	55	12%	18.2	2.2	1	50.7	13.2	
Exam 161G	FC-A-04	FCU	Note #1	Note #1	110	10.0	1,100	165	38	23%	9.0	2.1	1	58.5	6.6	
Exam 161H	FC-A-04	FCU	Note #1	Note #1	110	10.0	1,100	155	36	23%	8.5	1.9	1	58.1	6.2	
Exam 161J	FC-A-04	FCU	Note #1	Note #1	130	10.0	1,300	155	36	23%	7.2	1.6	1	49.1	5.3	
Choir 163	FC-A-05	FCU	Note #1	Note #1	1,534	14.0	21,476	1,460	642	44%	4.1	1.8	1	5.7	3.1	
Office 163A	FC-A-05	FCU	Note #1	Note #1	160	10.0	1,600	90	40	44%	3.4	1.5	1	38.2	2.5	
Practice 165E	FC-A-06	FCU	Note #1	Note #1	99	10.0	990	80	29	36%	4.8	1.7	1	61.2	3.6	
Practice 163B	FC-A-06	FCU	Note #1	Note #1	92	10.0	920	85	31	36%	5.5	2.0	1	66.1	4.1	
Practice 165D	FC-A-06	FCU	Note #1	Note #1	91	10.0	910	85	31	36%	5.6	2.0	1	66.8	4.2	
Practice 165C	FC-A-06	FCU	Note #1	Note #1	91	10.0	910	85	31	36%	5.6	2.0	1	66.8	4.2	

Date: 9/14/2021 Readings By: Jason Coutu

Where to learn more in McDaniel report: Page 12

District Airflow Measurements



Valuable, powerful data

Where to learn more in McDaniel report: Pages 14-20

Neudorfer Engineers. Inc.
Consulting Engineers Seattle Portland
www.NeudorferEngineers.com

AIRFLOW SURVEY REPORT

Project: Portland Public Schools Airflow Testing
Location: McDaniel HS; 2735 NE 82nd Ave., Portland, OR 97220
Filter Status: Upgraded

Room	Equipment Info		Room Dimensions					Airflow Measurements			Calculated ACH					Notes
	Served By	Equipment Type	Room Length	Room Width	Room Area	Room Height	Room Volume	Total CFM Supply	OA CFM Supply	OA %	Air Changes per Hour (supply)	Air Changes per Hour (OA)	# of Portable Filters	Total Effective Air Changes per Hour (ACH_e) with Portable Filter	Total Effective Air Changes per Hour (ACH_e) without Portable Filter	
Level 0																
Office 031D	FC-E-08	FCU	Note #1	Note #1	93	10.0	930	65	12	18%	4.2	0.8	1	64.4	3.1	
Office 031E	FC-E-08	FCU	Note #1	Note #1	98	10.0	980	65	12	18%	4.0	0.7	1	61.1	2.9	
Conf Rm 029	FC-E-06	FCU	Note #1	Note #1	236	8.8	2,084	245	76	31%	7.1	2.2	1	32.6	5.3	
Rm 027	FC-E-06	FCU	Note #1	Note #1	206	8.8	1,819	225	70	31%	7.4	2.3	1	36.9	5.5	
Rm 015	FC-E-02	FCU	Note #1	Note #1	646	10.0	6,460	370	303	82%	3.4	2.8	1	11.5	2.7	
Rm 013A	FC-E-01	FCU	Note #1	Note #1	979	10.0	9,790	950	460	48%	5.8	2.8	1	10.2	4.4	
013B	FC-E-03	FCU	Note #1	Note #1	978	10.0	9,780	925	450	49%	5.7	2.8	1	10.1	4.3	
ch 161	FC-A-03	FCU	Note #1	Note #1	153	10.0	1,530	255	26	10%	4.8	0.5	1	21.3	3.5	
on 161K	-	-	Note #1	Note #1	167	10.0	1,670	-	-	-						Supply air is provided by Health 161.
e 161A	FC-A-03	FCU	Note #1	Note #1	120	10.0	1,200	255	26	10%	12.8	1.3	1	56.8	9.3	
	A-03	FCU	Note #1	Note #1	161	10.0	1,610	250	25	10%	9.3	0.9	1	42.2	6.8	
	-A-03	FCU	Note #1	Note #1	121	10.0	1,210	230	23	10%	11.4	1.1	1	55.4	8.3	
	C-A-01	FCU	Note #1	Note #1	167	10.0	1,670	495	50	10%	17.8	1.8	1	47.1	12.9	
	FC-A-02	FCU	Note #1	Note #1	152	10.0	1,520	460	55	12%	18.2	2.2	1	50.7	13.2	
	FC-A-04	FCU	Note #1	Note #1	110	10.0	1,100	165	38	23%	9.0	2.1	1	58.5	6.6	
	FC-A-04	FCU	Note #1	Note #1	110	10.0	1,100	155	36	23%	8.5	1.9	1	58.1	6.2	
	C-A-04	FCU	Note #1	Note #1	130	10.0	1,300	155	36	23%	7.2	1.6	1	49.1	5.3	
		FCU	Note #1	Note #1	1,534	14.0	21,476	1,460	642	44%	4.1	1.8	1	5.7	3.1	
		FCU	Note #1	Note #1	160	10.0	1,600	90	40	44%	3.4	1.5	1	38.2	2.5	
		FCU	Note #1	Note #1	99	10.0	990	80	29	36%	4.8	1.7	1	61.2	3.6	
	C-A-06	FCU	Note #1	Note #1	92	10.0	920	85	31	36%	5.5	2.0	1	66.1	4.1	
	FC-A-06	FCU	Note #1	Note #1	91	10.0	910	85	31	36%	5.6	2.0	1	66.8	4.2	
	FC-A-06	FCU	Note #1	Note #1	91	10.0	910	85	31	36%	5.6	2.0	1	66.8	4.2	

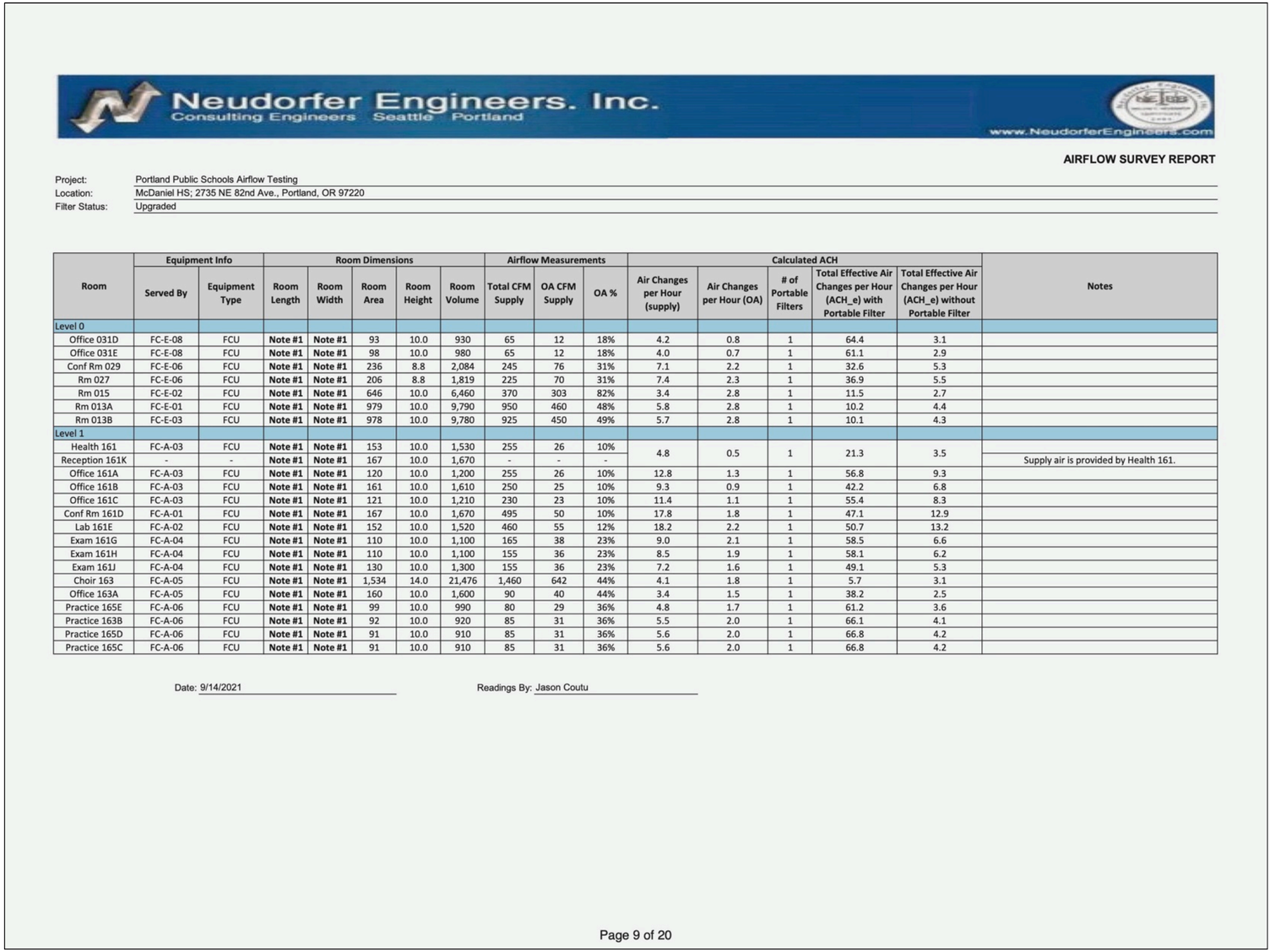
Date: 9/14/2021 Readings By: Jason Coutu

SIAFOS Database

- **Corrects errors**
- **Allows calculations of clean airflow**

* Air changes per hour
* Time to clear pollutants
* Total clean airflow or per person per minute
* Assess ASHRAE 62.1
* Assess ASHRAE 241
* Airflow added by any model of air purifier
* Expected CO_2

District Airflow Measurements



Neudorfer Engineers. Inc.
Consulting Engineers Seattle Portland
www.NeudorferEngineers.com

AIRFLOW SURVEY REPORT

Project: Portland Public Schools Airflow Testing
Location: McDaniel HS; 2735 NE 82nd Ave., Portland, OR 97220
Filter Status: Upgraded

Room	Equipment Info		Room Dimensions					Airflow Measurements			Calculated ACH					Notes
	Served By	Equipment Type	Room Length	Room Width	Room Area	Room Height	Room Volume	Total CFM Supply	OA CFM Supply	OA %	Air Changes per Hour (supply)	Air Changes per Hour (OA)	# of Portable Filters	Total Effective Air Changes per Hour (ACH_e) with Portable Filter	Total Effective Air Changes per Hour (ACH_e) without Portable Filter	
Level 0																
Office 031D	FC-E-08	FCU	Note #1	Note #1	93	10.0	930	65	12	18%	4.2	0.8	1	64.4	3.1	
Office 031E	FC-E-08	FCU	Note #1	Note #1	98	10.0	980	65	12	18%	4.0	0.7	1	61.1	2.9	
Conf Rm 029	FC-E-06	FCU	Note #1	Note #1	236	8.8	2,084	245	76	31%	7.1	2.2	1	32.6	5.3	
Rm 027	FC-E-06	FCU	Note #1	Note #1	206	8.8	1,819	225	70	31%	7.4	2.3	1	36.9	5.5	
Rm 015	FC-E-02	FCU	Note #1	Note #1	646	10.0	6,460	370	303	82%	3.4	2.8	1	11.5	2.7	
Rm 013A	FC-E-01	FCU	Note #1	Note #1	979	10.0	9,790	950	460	48%	5.8	2.8	1	10.2	4.4	
Rm 013B	FC-E-03	FCU	Note #1	Note #1	978	10.0	9,780	925	450	49%	5.7	2.8	1	10.1	4.3	
Level 1																
Health 161	FC-A-03	FCU	Note #1	Note #1	153	10.0	1,530	255	26	10%	4.8	0.5	1	21.3	3.5	
Reception 161K	-	-	Note #1	Note #1	167	10.0	1,670	-	-	-						Supply air is provided by Health 161.
Office 161A	FC-A-03	FCU	Note #1	Note #1	120	10.0	1,200	255	26	10%	12.8	1.3	1	56.8	9.3	
Office 161B	FC-A-03	FCU	Note #1	Note #1	161	10.0	1,610	250	25	10%	9.3	0.9	1	42.2	6.8	
Office 161C	FC-A-03	FCU	Note #1	Note #1	121	10.0	1,210	230	23	10%	11.4	1.1	1	55.4	8.3	
Conf Rm 161D	FC-A-01	FCU	Note #1	Note #1	167	10.0	1,670	495	50	10%	17.8	1.8	1	47.1	12.9	
Lab 161E	FC-A-02	FCU	Note #1	Note #1	152	10.0	1,520	460	55	12%	18.2	2.2	1	50.7	13.2	
Exam 161G	FC-A-04	FCU	Note #1	Note #1	110	10.0	1,100	165	38	23%	9.0	2.1	1	58.5	6.6	
Exam 161H	FC-A-04	FCU	Note #1	Note #1	110	10.0	1,100	155	36	23%	8.5	1.9	1	58.1	6.2	
Exam 161J	FC-A-04	FCU	Note #1	Note #1	130	10.0	1,300	155	36	23%	7.2	1.6	1	49.1	5.3	
Choir 163	FC-A-05	FCU	Note #1	Note #1	1,534	14.0	21,476	1,460	642	44%	4.1	1.8	1	5.7	3.1	
Office 163A	FC-A-05	FCU	Note #1	Note #1	160	10.0	1,600	90	40	44%	3.4	1.5	1	38.2	2.5	
Practice 165E	FC-A-06	FCU	Note #1	Note #1	99	10.0	990	80	29	36%	4.8	1.7	1	61.2	3.6	
Practice 163B	FC-A-06	FCU	Note #1	Note #1	92	10.0	920	85	31	36%	5.5	2.0	1	66.1	4.1	
Practice 165D	FC-A-06	FCU	Note #1	Note #1	91	10.0	910	85	31	36%	5.6	2.0	1	66.8	4.2	
Practice 165C	FC-A-06	FCU	Note #1	Note #1	91	10.0	910	85	31	36%	5.6	2.0	1	66.8	4.2	

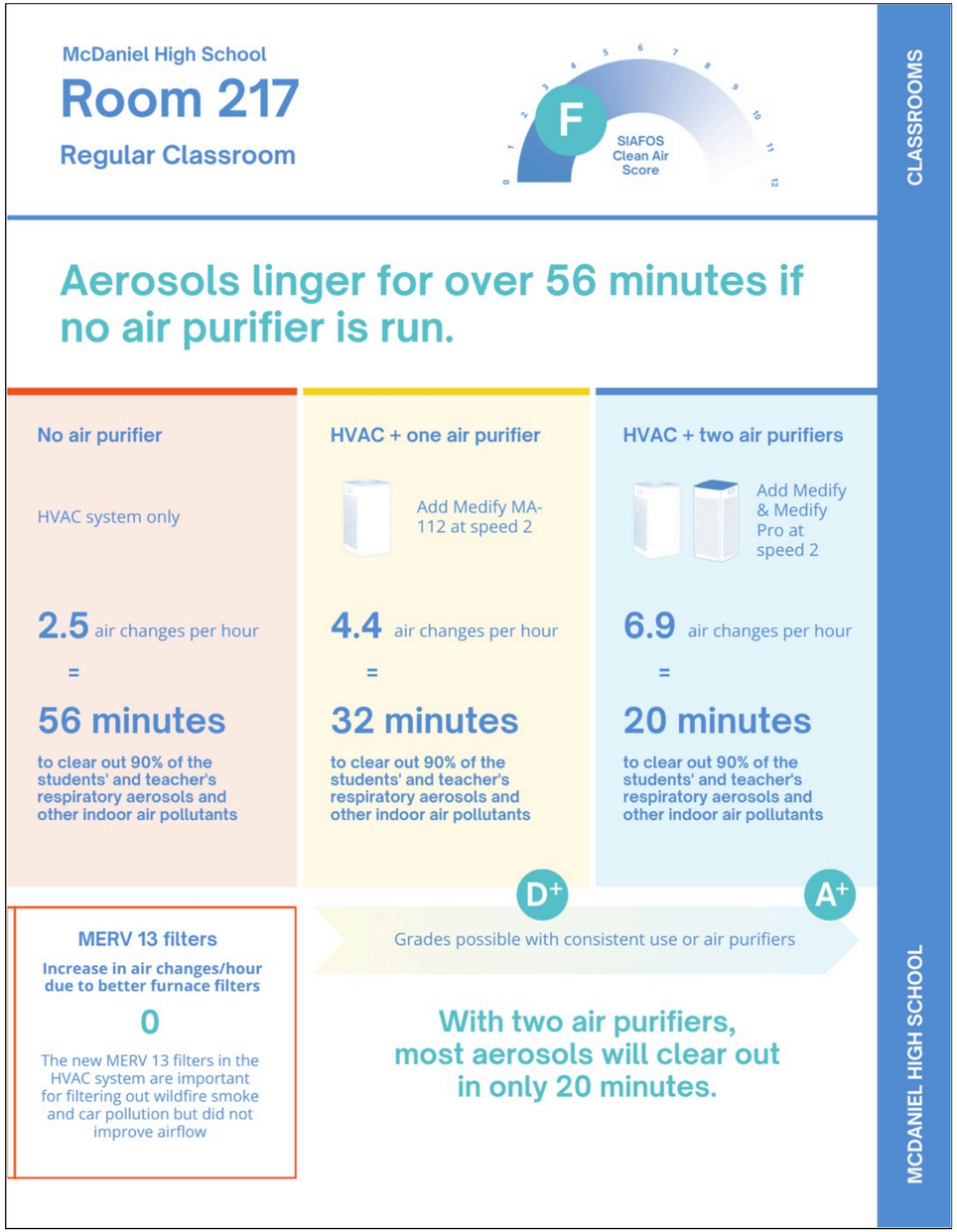
Date: 9/14/2021 Readings By: Jason Coutu

Page 9 of 20

Room-by-Room Air Quality Profiles

Where to learn more in McDaniel report: Pp. 48-129



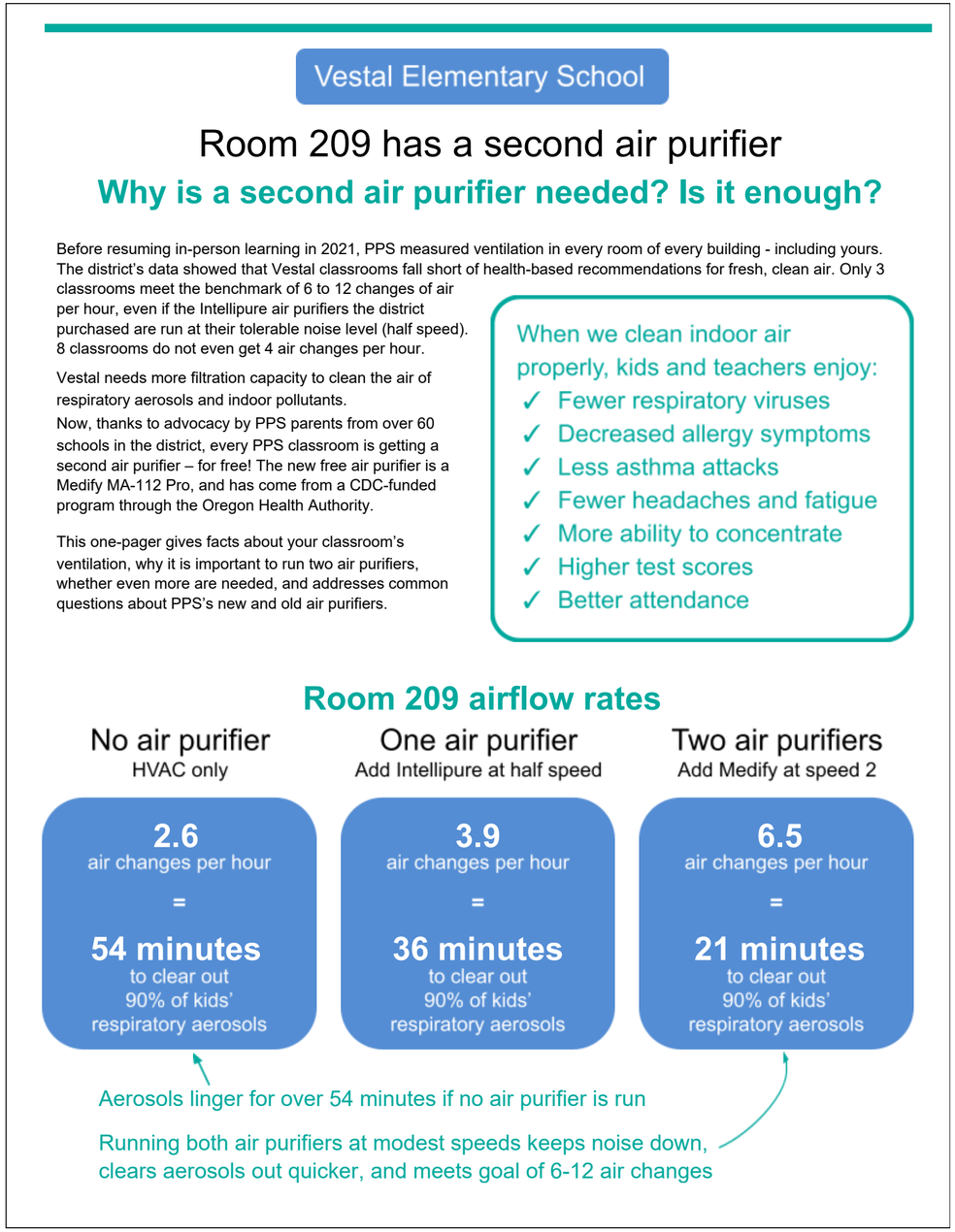


Semi-automated system to produce:

Recommended air purifier placements by room and by school

Custom one-pagers for teachers

Where to learn more in McDaniel report: Pp 43-45



Vestal Elementary School

Room 209 has a second air purifier

Why is a second air purifier needed? Is it enough?

Before resuming in-person learning in 2021, PPS measured ventilation in every room of every building - including yours. The district's data showed that Vestal classrooms fall short of health-based recommendations for fresh, clean air. Only 3 classrooms meet the benchmark of 6 to 12 changes of air per hour, even if the Intellipure air purifiers the district purchased are run at their tolerable noise level (half speed). 8 classrooms do not even get 4 air changes per hour.

Vestal needs more filtration capacity to clean the air of respiratory aerosols and indoor pollutants.

Now, thanks to advocacy by PPS parents from over 60 schools in the district, every PPS classroom is getting a second air purifier – for free! The new free air purifier is a Medify MA-112 Pro, and has come from a CDC-funded program through the Oregon Health Authority.

This one-pager gives facts about your classroom's ventilation, why it is important to run two air purifiers, whether even more are needed, and addresses common questions about PPS's new and old air purifiers.

When we clean indoor air properly, kids and teachers enjoy:

- ✓ Fewer respiratory viruses
- ✓ Decreased allergy symptoms
- ✓ Less asthma attacks
- ✓ Fewer headaches and fatigue
- ✓ More ability to concentrate
- ✓ Higher test scores
- ✓ Better attendance

Room 209 airflow rates

No air purifier	One air purifier	Two air purifiers
HVAC only	Add Intellipure at half speed	Add Medify at speed 2
2.6 air changes per hour = **54 minutes** to clear out 90% of kids' respiratory aerosols	**3.9** air changes per hour = **36 minutes** to clear out 90% of kids' respiratory aerosols	**6.5** air changes per hour = **21 minutes** to clear out 90% of kids' respiratory aerosols

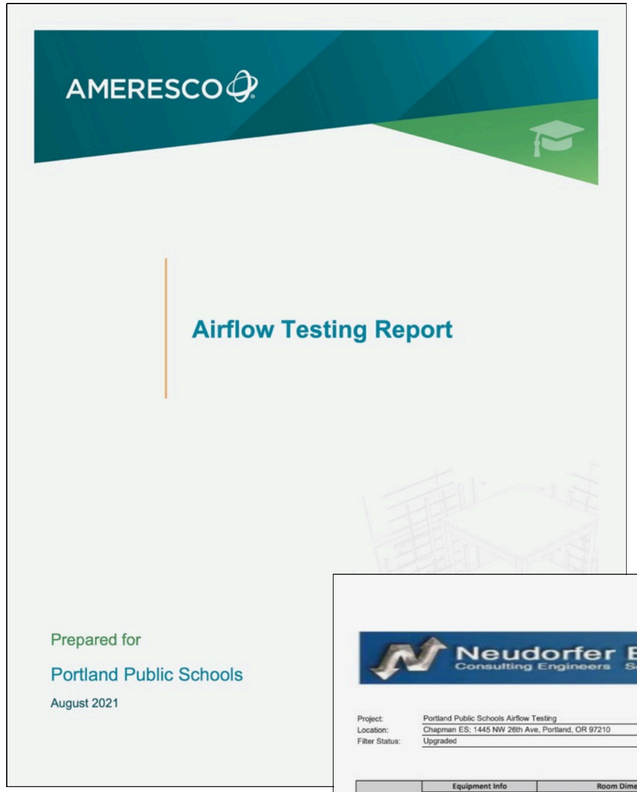
Aerosols linger for over 54 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and meets goal of 6-12 air changes

Chapman Elementary School

Where to learn more in McDaniel report: Page 3 & 37-42



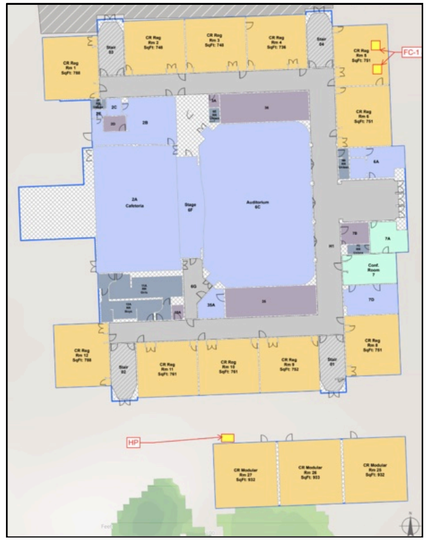

AMERESCO

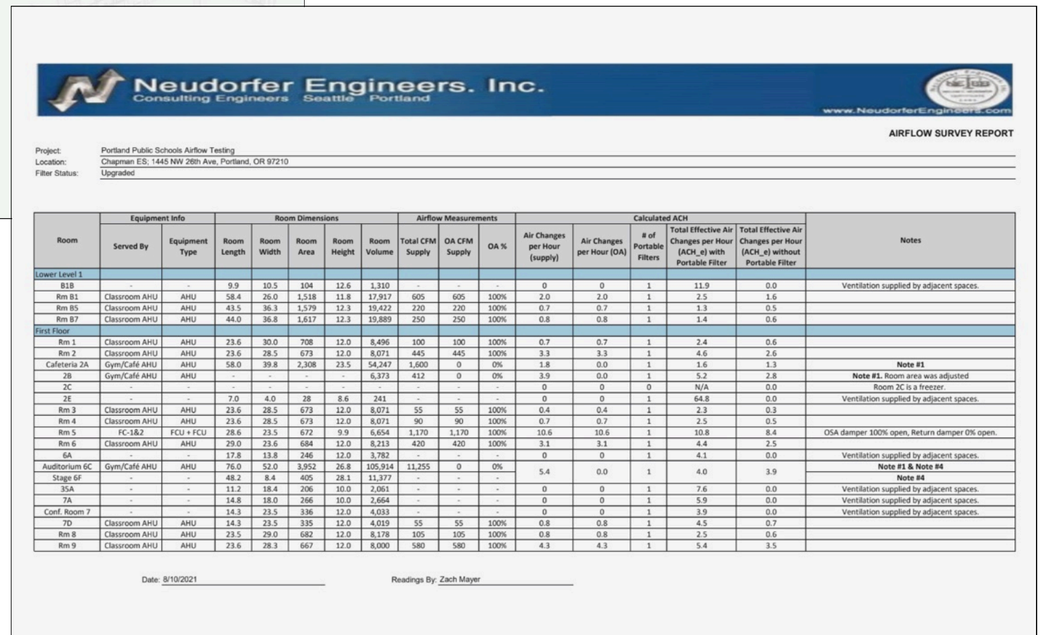
Airflow Testing Report

Prepared for

Portland Public Schools

August 2021




Neudorfer Engineers, Inc.
Consulting Engineers Seattle Portland
www.NeudorferEngineers.com

AIRFLOW SURVEY REPORT

Project: Portland Public Schools Airflow Testing
Location: Chapman ES; 1445 NW 26th Ave, Portland, OR 97210
Filter Status: Upgraded

Room	Equipment Info		Room Dimensions					Airflow Measurements			Calculated ACH				Notes
	Served By	Equipment Type	Room Length	Room Width	Room Area	Room Height	Room Volume	Total CFM Supply	OA CFM Supply	OA %	Air Changes per Hour (supply)	Air Changes per Hour (OA)	# of Portable Filters	Total Effective Air Changes per Hour (ACH_e) with Portable Filter	Total Effective Air Changes per Hour (ACH_e) without Portable Filter	
Lower Level 1																
B1B	-	-	9.9	10.5	104	12.6	1,310	-	-	-	0	0	1	11.9	0.0	Ventilation supplied by adjacent spaces.
Rm B1	Classroom AHU	AHU	58.4	26.0	1,518	11.8	17,917	605	605	100%	2.0	2.0	1	2.5	1.6	
Rm B5	Classroom AHU	AHU	43.5	36.3	1,579	12.3	19,422	220	220	100%	0.7	0.7	1	1.3	0.5	
Rm B7	Classroom AHU	AHU	44.0	36.8	1,617	12.3	19,889	250	250	100%	0.8	0.8	1	1.4	0.6	
First Floor																
Rm 1	Classroom AHU	AHU	23.6	30.0	708	12.0	8,496	100	100	100%	0.7	0.7	1	2.4	0.6	
Rm 2	Classroom AHU	AHU	23.6	28.5	673	12.0	8,071	445	445	100%	3.3	3.3	1	4.6	2.6	
Cafeteria 2A	Gym/Café AHU	AHU	58.0	39.8	2,308	23.5	54,247	1,600	0	0%	1.8	0.0	1	1.6	1.3	**Note #1**
2B	Gym/Café AHU	AHU	-	-	-	-	6,373	412	0	0%	3.9	0.0	1	5.2	2.8	**Note #1.** Room area was adjusted
2C	-	-	-	-	-	-	-	-	-	-	0	0	0	N/A	0.0	Room 2C is a freezer.
2E	-	-	7.0	4.0	28	8.6	241	-	-	-	0	0	1	64.8	0.0	Ventilation supplied by adjacent spaces.
Rm 3	Classroom AHU	AHU	23.6	28.5	673	12.0	8,071	55	55	100%	0.4	0.4	1	2.3	0.3	
Rm 4	Classroom AHU	AHU	23.6	28.5	673	12.0	8,071	90	90	100%	0.7	0.7	1	2.5	0.5	
Rm 5	FC-1&2	FCU + FCU	28.6	23.5	672	9.9	6,654	1,170	1,170	100%	10.6	10.6	1	10.8	8.4	OSA damper 100% open, Return damper 0% open.
Rm 6	Classroom AHU	AHU	29.0	23.6	684	12.0	8,213	420	420	100%	3.1	3.1	1	4.4	2.5	
6A	-	-	17.8	13.8	246	12.0	3,782	-	-	-	0	0	1	4.1	0.0	Ventilation supplied by adjacent spaces.
Auditorium 6C	Gym/Café AHU	AHU	76.0	52.0	3,952	26.8	105,914	11,255	0	0%	5.4	0.0	1	4.0	3.9	**Note #1 & Note #4**
Stage 6F	-	-	48.2	8.4	405	28.1	11,377	-	-	-						**Note #4**
35A	-	-	11.2	18.4	206	10.0	2,061	-	-	-	0	0	1	7.6	0.0	Ventilation supplied by adjacent spaces.
7A	-	-	14.8	18.0	266	10.0	2,664	-	-	-	0	0	1	5.9	0.0	Ventilation supplied by adjacent spaces.
Conf. Room 7	-	-	14.3	23.5	336	12.0	4,033	-	-	-	0	0	1	3.9	0.0	Ventilation supplied by adjacent spaces.
7D	Classroom AHU	AHU	14.3	23.5	335	12.0	4,019	55	55	100%	0.8	0.8	1	4.5	0.7	
Rm 8	Classroom AHU	AHU	23.5	29.0	682	12.0	8,178	105	105	100%	0.8	0.8	1	2.5	0.6	
Rm 9	Classroom AHU	AHU	23.6	28.3	667	12.0	8,000	580	580	100%	4.3	4.3	1	5.4	3.5	

Date: 8/10/2021 Readings By: Zach Mayer

- An example that illustrates how the McDaniel Report is a case study for district-wide issues with air quality
- Title 1 school: 52% direct certification
- 41% historically underserved
- Long-term concern: air toxics from industry

Chapman Elementary School Airflow Report

AMERESCO

Airflow Testing

Prepared for

Portland Public Schools

August 2021

Neudorfer Engineers. Inc.
Consulting Engineers Seattle Portland
www.NeudorferEngineers.com

AIRFLOW SURVEY REPORT

Project: Portland Public Schools Airflow Testing
Location: Chapman ES; 1445 NW 26th Ave, Portland, OR 97210
Filter Status: Upgraded

Room	Equipment Info		Room Dimensions					Airflow Measurements			Calculated ACH					Notes
	Served By	Equipment Type	Room Length	Room Width	Room Area	Room Height	Room Volume	Total CFM Supply	OA CFM Supply	OA %	Air Changes per Hour (supply)	Air Changes per Hour (OA)	# of Portable Filters	Total Effective Air Changes per Hour (ACH_e) with Portable Filter	Total Effective Air Changes per Hour (ACH_e) without Portable Filter	
Lower Level 1																
B1B	-	-	9.9	10.5	104	12.6	1,310	-	-	-	0	0	1	11.9	0.0	Ventilation supplied by adjacent spaces.
Rm B1	Classroom AHU	AHU	58.4	26.0	1,518	11.8	17,917	605	605	100%	2.0	2.0	1	2.5	1.6	
Rm B5	Classroom AHU	AHU	43.5	36.3	1,579	12.3	19,422	220	220	100%	0.7	0.7	1	1.3	0.5	
Rm B7	Classroom AHU	AHU	44.0	36.8	1,617	12.3	19,889	250	250	100%	0.8	0.8	1	1.4	0.6	
First Floor																
Rm 1	Classroom AHU	AHU	23.6	30.0	708	12.0	8,496	100	100	100%	0.7	0.7	1	2.4	0.6	
Rm 2	Classroom AHU	AHU	23.6	28.5	673	12.0	8,071	445	445	100%	3.3	3.3	1	4.6	2.6	
Cafeteria 2A	Gym/Café AHU	AHU	58.0	39.8	2,308	23.5	54,247	1,600	0	0%	1.8	0.0	1	1.6	1.3	**Note #1**
2B	Gym/Café AHU	AHU	-	-	-	-	6,373	412	0	0%	3.9	0.0	1	5.2	2.8	**Note #1.** Room area was adjusted
2C	-	-	-	-	-	-	-	-	-	-	0	0	0	N/A	0.0	Room 2C is a freezer.
2E	-	-	7.0	4.0	28	8.6	241	-	-	-	0	0	1	64.8	0.0	Ventilation supplied by adjacent spaces.
Rm 3	Classroom AHU	AHU	23.6	28.5	673	12.0	8,071	55	55	100%	0.4	0.4	1	2.3	0.3	
Rm 4	Classroom AHU	AHU	23.6	28.5	673	12.0	8,071	90	90	100%	0.7	0.7	1	2.5	0.5	
Rm 5	FC-1&2	FCU + FCU	28.6	23.5	672	9.9	6,654	1,170	1,170	100%	10.6	10.6	1	10.8	8.4	OSA damper 100% open, Return damper 0% open.
Rm 6	Classroom AHU	AHU	29.0	23.6	684	12.0	8,213	420	420	100%	3.1	3.1	1	4.4	2.5	
6A	-	-	17.8	13.8	246	12.0	3,782	-	-	-	0	0	1	4.1	0.0	Ventilation supplied by adjacent spaces.
Auditorium 6C	Gym/Café AHU	AHU	76.0	52.0	3,952	26.8	105,914	11,255	0	0%	5.4	0.0	1	4.0	3.9	**Note #1 & Note #4**
Stage 6F	-	-	48.2	8.4	405	28.1	11,377	-	-	-						**Note #4**
35A	-	-	11.2	18.4	206	10.0	2,061	-	-	-	0	0	1	7.6	0.0	Ventilation supplied by adjacent spaces.
7A	-	-	14.8	18.0	266	10.0	2,664	-	-	-	0	0	1	5.9	0.0	Ventilation supplied by adjacent spaces.
Conf. Room 7	-	-	14.3	23.5	336	12.0	4,033	-	-	-	0	0	1	3.9	0.0	Ventilation supplied by adjacent spaces.
7D	Classroom AHU	AHU	14.3	23.5	335	12.0	4,019	55	55	100%	0.8	0.8	1	4.5	0.7	
Rm 8	Classroom AHU	AHU	23.5	29.0	682	12.0	8,178	105	105	100%	0.8	0.8	1	2.5	0.6	
Rm 9	Classroom AHU	AHU	23.6	28.3	667	12.0	8,000	580	580	100%	4.3	4.3	1	5.4	3.5	

Date: 8/10/2021 Readings By: Zach Mayer



Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Air changes per hour here are effective air changes per hour = air change rates from outdoor airflow + clean airflow from recirculation through MERV 13 filters

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	[illegible]	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Classroom 1

0.6 air changes/hour if no air purifiers running

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	[illegible]	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	[illegible]	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Classroom 2

2.6 air changes/hour if no air purifiers running

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	[illegible]	1	2.5
Rm 9	3.5	1	5.4
Rm 10	[illegible]	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Classroom 9

3.5 air changes/hour if no air purifiers running

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Second Floor			
Rm 13	2.7	1	4.6
Rm 14	2.2	1	4.1
33	0.0	1	59.6
Rm 15	0.5	1	2.5
Rm 16	2.1	1	4.0
Library 17	2.8	1	3.7
18	1.3	1	2.6
18B	0.0	0	N/A
Balcony 31	0.0	1	1.8
29	0.0	0	N/A
Rm 19	1.3	1	3.2
Rm 20	2.9	1	4.8
Rm 21	2.2	1	4.1
Rm 22	0.6	1	2.6
Rm 23	0.4	1	2.3
Rm 24	2.4	1	4.3
Rm 25	4.9	1	6.7
Rm 26	4.3	1	6.1
Rm 27	4.3	1	6.1

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Second Floor			
Rm 13	2.7	1	4.6
Rm 14	2.2	1	4.1
33	0.0	1	59.6
Rm 15	0.5	1	2.5
Rm 16	2.1	1	4.0
Library 17	2.8	1	3.7
18	1.3	1	2.6
18B	0.0	0	N/A
Balcony 31	0.0	1	1.8
29	0.0	0	N/A
Rm 19	1.3	1	3.2
Rm 20	2.9	1	4.8
Rm 21	2.2	1	4.1
Rm 22	0.6	1	2.6
Rm 23	0.4	1	2.3
Rm 24	2.4	1	4.3
Rm 25	4.9	1	6.7
Rm 26	4.3	1	6.1
Rm 27	4.3	1	6.1

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0



Total air changes per hour with Intellipure at full “turbo” speed

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	[illegible]
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Classroom 1

2.4 air changes/hour if teacher runs the Intellipure air purifier at full speed

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Classroom 2

4.6 air changes/hour if teacher runs the Intellipure air purifier at full speed

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Classroom 2

4.6 air changes/hour if teacher runs the Intellipure air purifier at full speed

59 decibels

<u>TOO NOISY FOR CLASSROOMS</u>

Where to learn more in McDaniel report: Page 29

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Where to learn more in McDaniel report: Page 19

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**



Total air changes per hour with Intellipure at half speed

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	[illegible]	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	[illegible]	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Classroom 1

1.9 air changes/hour with one air purifier

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	[illegible]	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	[illegible]	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Classroom 2

4 air changes/hour with one air purifier

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Most PPS schools have these kinds of low amounts of clean air in classrooms with only one air purifier



Español | Tiếng Việt | 中文 | Русский | Soomaali

Our Continuing Commitment to Air Quality

May 2, 2023

Dear Portland Public School families,

We strive to ensure that all or your student's learning spaces are inspiring, joyful, and safe. Air quality is one element of school safety, and we provided an update in October about how we maximize the air quality in our schools. This includes:

- Following the consensus advice of public health experts
- Meeting public health authority regulations
- Partnering with PBS Engineering and Environmental, one of the Northwest's top environmental engineering firms
- Continuous review and improvement with PBS, public health experts, and our community

Over the last week, our commitment to continuous review was on display as multiple PPS offices came together to review our systems. Our review aimed to take advantage of an Oregon Health Authority program to grow our inventory of filters and air purifiers at no cost to the district. Our teams took feedback from our schools, central operations, and community members, and placed an order for 3,500 new air purifiers. We also anticipate getting a five-year supply of filters for those 3,500 purifiers – AND nearly all of our existing 5464 purifiers.

As a result, we anticipate each PPS classroom will have two air purifiers sharing the work to cleanse the air and boost the number of air changes per hour. This is a crucial part of our layered-mitigation strategy:

Where to learn more in McDaniel report: Pp 37-39

PORTLAND PUBLIC SCHOOLS
501 North Dixon Street
Portland, OR 97227
(503) 916-2000

Memo – Air Purifier Distribution Plan

To: Principals

Date: August 31, 2023

The District has begun to receive the 3,500 Medify air purifiers and a five-year supply of filter replacements from the Oregon Health Authority (OHA). These Medify air purifiers will be distributed to 28 of our schools with classrooms that are on the lower end of our *air changes per hour* (ACH) measurements. Additional units can be provided upon request to all schools.

All PPS schools already have healthy air quality
Air quality in all PPS schools is safe. We follow consensus advice of public health experts to meet all indoor air quality regulations. PPS also partners with top air quality experts to ensure we meet industry best practices. In 2021, an independent consulting firm, PBS Engineering and Environmental completed air quality assessments at all PPS schools and determined indoor air quality is good at all PPS schools. Copies of these reports can be found on the Environmental Health & Safety website.

Airflow is one component of indoor air quality
One component of indoor air quality is airflow, often measured as **Air Changes per Hour (ACH)**. Airflow changes throughout the day and is based on a number of variables. Please note - there is no common standard for how to measure airflow. To gather airflow information about each PPS school, the district partnered with Ameresco to complete *one point in time testing* during the 2021 and 2022 school year. This information gives us a generalized sense of which schools have generally higher ACH and which ones are lower. These ACH reports can be found on the Environmental Health & Safety website.

The District is providing additional air purifiers to specific school classrooms
While there is no standard for airflow in schools, the Oregon Health Authority recommends 3 to 6 air changes per hour. Currently all PPS classrooms are supplied with one portable air purifier. Our review of the ACH data shows that at 68 schools, classrooms average more than 6 ACH (when the portable air purifier is turned on) while 28 schools' classrooms fall below that average. We are providing one additional unit for each classroom in those 28 schools with the understanding some schools or spaces may also want additional airflow.

Additional air purifiers can be provided upon request to all schools
The site administrator may request additional units by emailing tenbysk@pps.net. Please gather the entire request amount for your school before you submit your request so we only deliver to your school once.

The 28 schools that will be receiving additional units are:

1. Arleta
2. Astor
3. Atkinson
4. Beach
5. Boise-Eliot
6. Bridger
7. Buckman
8. Clark
9. East Sylvan
10. Edwards
11. Glencoe
12. Hosford
13. King
14. KairosPDX
15. Laurelhurst
16. Lee
17. Lent
18. Rieke
19. Sabin
20. Sellwood
21. Skyline
22. Sunnyside
23. Terwilliger
24. Vestal
25. West Sylvan
26. Winterhaven
27. Woodlawn
28. Woodstock

While air quality in all PPS schools remains safe, the District is committed to providing additional improvement opportunities. As schools or individual air handling systems are modernized in the upcoming years, the District will reassess with an eye to lowering energy use and benefiting the PPS climate policy efforts.

Air Purifier Maintenance
The District has purchased two (2) different types of air purifiers. The first is the Intellipure Ultrafine 468 unit, and the second is the MedifyAir MA112 unit. Both units require regular cleaning and

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Average was of all rooms in the report, not just classrooms

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
[illegible]	0.0	0	N/A
2E	0.0	1	64.8
[illegible]	0.3	1	[illegible]
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Room 2E = storage closet in kitchen with a claim of 64.8 air changes per hour

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Lower Level 1			
B1B	0.0	1	11.9
Rm B1	1.6	1	2.5
Rm B5	0.5	1	1.3
Rm B7	0.6	1	1.4
First Floor			
Rm 1	0.6	1	2.4
Rm 2	2.6	1	4.6
Cafeteria 2A	1.3	1	1.6
2B	2.8	1	5.2
2C	0.0	0	N/A
2E	0.0	1	64.8
Rm 3	0.3	1	2.3
Rm 4	0.5	1	2.5
Rm 5	8.4	1	10.8
Rm 6	2.5	1	4.4
6A	0.0	1	4.1
Auditorium 6C Stage 6F	3.9	1	4.0
35A	0.0	1	7.6
7A	0.0	1	5.9
Conf. Room 7	0.0	1	3.9
7D	0.7	1	4.5
Rm 8	0.6	1	2.5
Rm 9	3.5	1	5.4
Rm 10	1.9	1	3.8
Rm 11	0.6	1	2.5
Rm 12	4.7	1	9.0

Room	Air Changes per Hour without Portable Filter	# of Portable Filters	Air Changes per Hour with Portable Filter
Second Floor			
Rm 13	2.7	1	4.6
Rm 14	2.2	1	4.1
33	0.0	1	59.6
Rm 15	0.5	1	2.5
Rm 16	2.1	1	4.0
Library 17	2.8	1	3.7
18	1.3	1	2.6
18B	0.0	0	N/A
Balcony 31	0.0	1	1.8
29	0.0	0	N/A
Rm 19	1.3	1	3.2
Rm 20	2.9	1	4.8
Rm 21	2.2	1	4.1
Rm 22	0.6	1	2.6
Rm 23	0.4	1	2.3
Rm 24	2.4	1	4.3
Rm 25	4.9	1	6.7
Rm 26	4.3	1	6.1
Rm 27	4.3	1	6.1

Average of classrooms only with Intellipure at full speed: 4.1 air changes/hour

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Second Floor			
Rm 13	4.0	6.8	**
Rm 14	3.5	6.4	**
33	10.3	32.2	?
Rm 15	1.9	4.7	**
Rm 16	3.5	6.4	**
Library 17	3.5	4.9	*
18	2.2	4.1	?
18B	0.0	11.3	?
Balcony 31	1.3	4.0	?
29	0.0	11.5	?
Rm 19	2.6	5.5	**
Rm 20	4.2	7.1	**
Rm 21	3.5	6.4	**
Rm 22	1.9	4.8	**
Rm 23	1.8	4.6	**
Rm 24	3.7	6.4	**
Rm 25	6.2	9.1	**
Rm 26	5.6	8.5	**
Rm 27	5.7	8.7	**

Realistic average of classrooms Intellipure at half speed: 3.5 air changes/hour

More examples of among schools not on the list:

	With one original air purifier*	
	Average	Lowest classroom
César Chávez	3.7	0.8
Faubion	2.9	0.5
George	4.2	1.1
Markham	2.7	1.0
McDaniel	5.5	1.9
Roosevelt	3.8	1.2
Rosa Parks	2.6	0.9
Tubman	3.6	0.9
Woodmere	2.7	1.0

Among elementary, middle, K-8 and high schools with a valid airflow report, only three actually have an average classroom airflow above 6 effective air changes per hour

*Using more accurately measured clean air delivery rates

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**



Total air changes per hour with two air purifiers at quiet speed levels

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	[illegible]	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Classroom 1

4.6 air changes/hour

if teacher runs an Intellipure air purifier at half speed & a Medify Pro at speed 2

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	
First Floor			
Rm 1	1.9	[illegible]	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	[illegible]	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	?
7D	3.4	9.2	?
Rm 8	1.9	4.8	**
Rm 9	4.9	7.8	**
Rm 10	3.2	6.1	**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Classroom 2

6.8 air changes/hour

if teacher runs an Intellipure air purifier at half speed & a Medify Pro at speed 2

Where to learn more in McDaniel report: Pp. 28-29

Room	Air Changes per Hour with Portable Filter at half speed	Air Changes per Hour with Portable Filter at half speed + Medify at speed 2	
Lower Level 1			
B1B	8.3	26.1	?
Rm B1	2.2	3.5	**
Rm B5	1.1	2.3	**
Rm B7	1.1	2.3	**
First Floor			
Rm 1	1.9	4.6	**
Rm 2	4.0	6.8	**
Cafeteria 2A	1.5	1.9	*
2B	4.5	8.2	?
2C	-	-	
2E	45.3	141.8	?
Rm 3	1.7	4.5	**
Rm 4	1.9	4.7	**
Rm 5	10.0	13.5	**
Rm 6	3.8	6.7	**
6A	3.7	11.6	?
Auditorium 6C	4.0	4.2	*
Stage 6F			
35A	5.3	16.6	?
7A	4.1	12.8	?
Conf. Room 7	2.7	8.5	
7D	3.4	9.2	?
Rm 8	1.9		**
Rm 9	4.9	7.8	**
Rm 10	3.2		**
Rm 11	2.0	4.8	**
Rm 12	7.7	13.9	**

Classroom 9

7.8 air changes/hour

if teacher runs an Intellipure air purifier at half speed & a Medify Pro at speed 2

Chapman Elementary School

Room 1 needs a second air purifier

Why is a second air purifier needed? Is it enough?

Before resuming in-person learning in 2021, PPS measured ventilation in every room of every building - including yours. The district's data showed that Chapman's classrooms fall short of health-based recommendations for fresh, clean air. Only 3 classrooms meet the benchmark of 6 to 12 changes of air per hour, even if the Intellipure air purifiers the district purchased are run at their tolerable noise level (half speed). 12 classrooms do not even get 3 air changes per hour.

Chapman needs more filtration capacity to clean the air of respiratory aerosols and indoor pollutants.

Now, thanks to advocacy by PPS parents from over 60 schools in the district, every PPS classroom is getting a second air purifier – for free! The new free air purifier is a Medify MA-112 Pro, and has come from a CDC-funded program through the Oregon Health Authority.

This one-pager gives facts about your classroom's ventilation, why it is important to run two air purifiers, whether even more are needed, and addresses common questions about PPS's new and old air purifiers.

When we clean indoor air properly, kids and teachers enjoy:

- ✓ Fewer respiratory viruses
- ✓ Decreased allergy symptoms
- ✓ Less asthma attacks
- ✓ Fewer headaches and fatigue
- ✓ More ability to concentrate
- ✓ Higher test scores
- ✓ Better attendance

Room 1 airflow rates

No air purifier HVAC only	One air purifier Add Intellipure at half speed	Two air purifiers Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Where to learn more in McDaniel report: Page 23

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

No air purifier HVAC only	One air purifier Add Intellipure at half speed	Two air purifiers Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols
respiratory aerosols
respiratory aerosols
Aerosols linger for over 4 hours if no air purifier is run
Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

No air purifier HVAC only	One air purifier Add Intellipure at half speed	Two air purifiers Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols
respiratory aerosols
respiratory aerosols
Aerosols linger for over 4 hours if no air purifier is run
Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

No air purifier	One air purifier	Two air purifiers
HVAC only	Add Intellipure at half speed	Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols respiratory aerosols respiratory aerosols

Aerosols linger for over 4 hours if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

No air purifier	One air purifier	Two air purifiers
HVAC only	Add Intellipure at hal[illegible]	Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols respiratory aerosols respiratory aerosols

Aerosols linger for over 4 hours if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

No air purifier	One air purifier	Two air purifiers
HVAC only	Add Intellipure at half speed	Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols respiratory aerosols respiratory aerosols

Aerosols linger for over 4 hours if no air purifier is run

Running both air purifiers at modest speeds keeps noise down, clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

No air purifier HVAC only	One air purifier Add Intellipure at half speed	Two air purifiers Add Medify at speed 1
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols
respiratory aerosols
respiratory aerosols
Aerosols linger for over 4 hours if no air purifier is run
Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

Chapman Elementary

Chapman Elementary School

Room 1 needs a second air purifier

Room 1 airflow rates

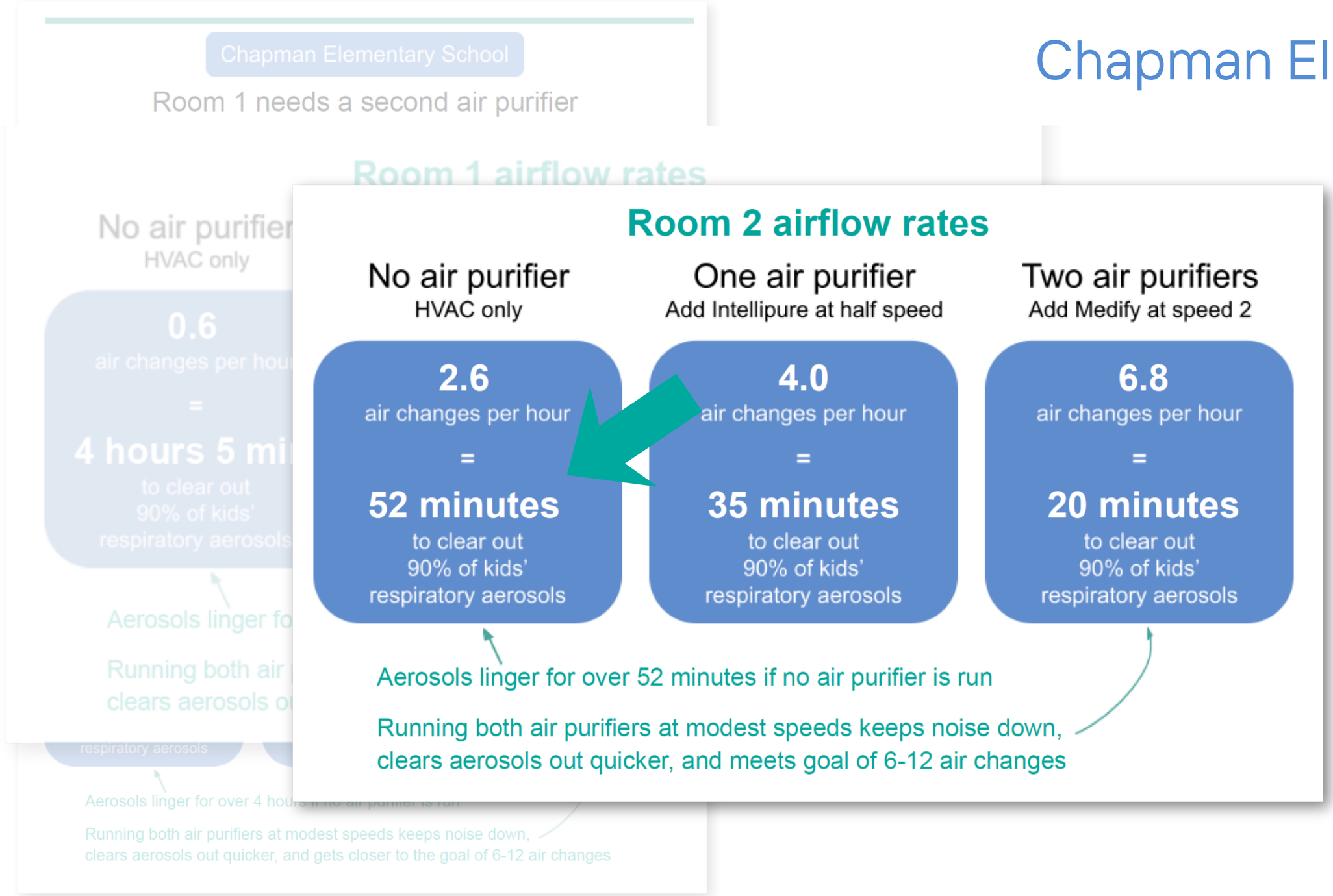
No air purifier HVAC only	One air purifier Add Intellipure at half speed	Two air purifiers Add Medify at speed 2
0.6 air changes per hour = **4 hours 5 min** to clear out 90% of kids' respiratory aerosols	**1.9** air changes per hour = **1 hour 13 min** to clear out 90% of kids' respiratory aerosols	**4.6** air changes per hour = **30 minutes** to clear out 90% of kids' respiratory aerosols

Aerosols linger for over 4 hours 5 minutes if no air purifier is run

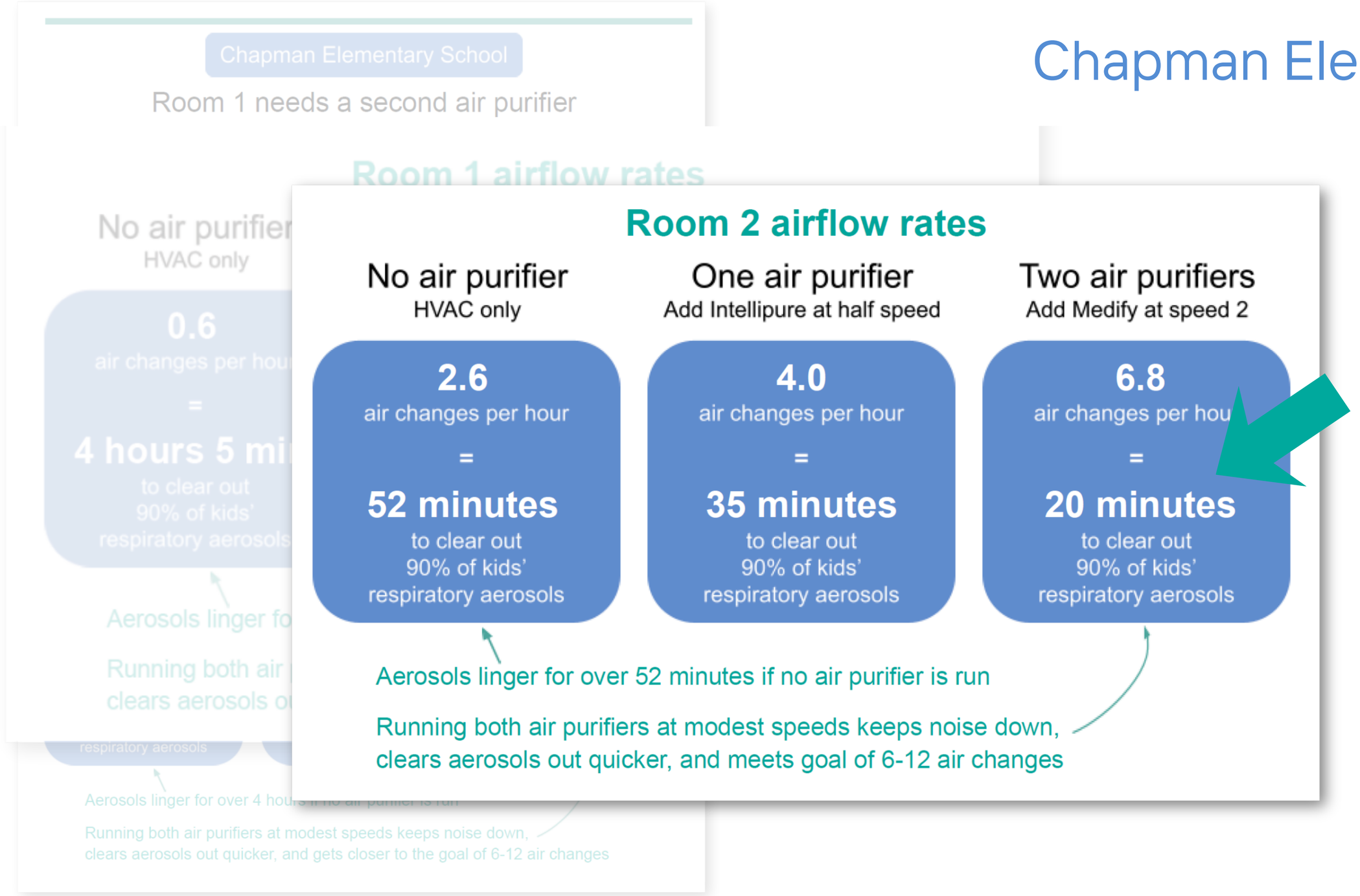
Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

respiratory aerosols
respiratory aerosols
respiratory aerosols
Aerosols linger for over 4 hours if no air purifier is run
Running both air purifiers at modest speeds keeps noise down,
clears aerosols out quicker, and gets closer to the goal of 6-12 air changes

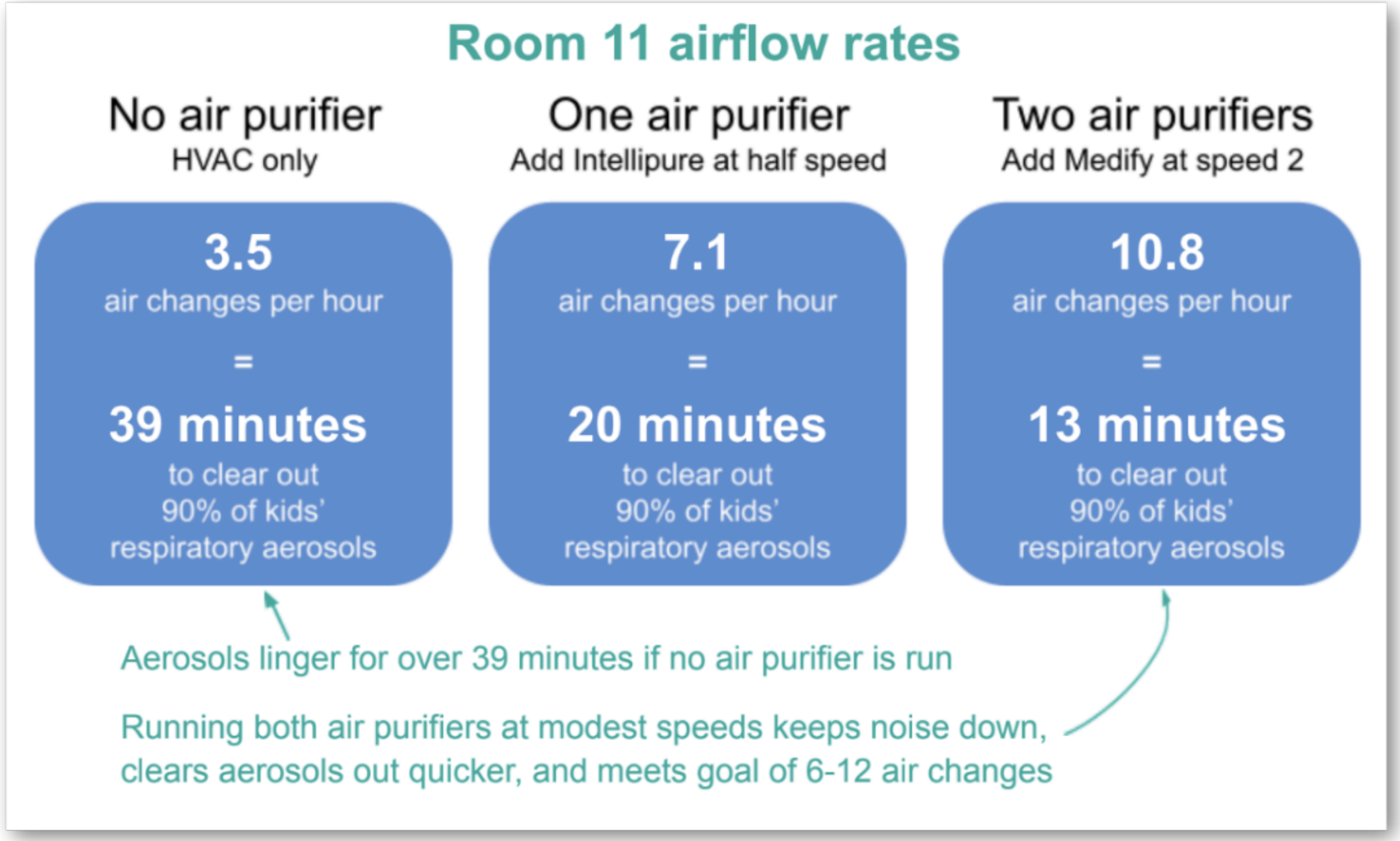
Chapman Elementary



Chapman Elementary



At other many schools, classrooms are achieving 10-12 effective air changes per hour



Benefits of clean indoor air



Bailey et al. 2022



Where to learn more in McDaniel report: Pp. 6-7

Benefits of clean indoor air

Less exposure to pollutants, triggers

Prevent new cases of asthma

Prevent asthma attacks



Where to learn more in McDaniel report: Page 7

Health risks when we fail to clean indoor air



Lewis Elementary during 2020 wildfires



Where to learn more in McDaniel report: Pp 8-9

Benefits of clean indoor air



Fresh, clean air

Fewer absences & sick days

Better focus & fewer errors

Improved student and staff performance

Plus fewer substitutes



Where to learn more in McDaniel report: Page 6

César Chávez K-8 School



- Title 1 school: 66% direct certification
- 82% historically underserved
- 36% multi-language learner

With only one air purifier per room, half of César Chávez classrooms are less than 3 air changes/hour

With two air purifier per classroom, no classrooms are less than 3 air change/hour & over half of classrooms are above our minimum goal

- Over $20,000 worth of air purifiers bought for them by OHA in 2023
- With SIAFOS support, requested their share of units – to cover classrooms, library, cafeteria, kitchen, and main office
- Received their units last week

Costs of clean air vs costs of not cleaning air

Where to learn more in McDaniel report: Page 136

	K-8 schools		High schools	
	1st air purifier: Intellipure	2nd air purifier: Medify Pro	1st air purifier: Regular Medify	2nd air purifier: Medify Pro
				
Filter changes	6 school years before filter change needed	At least 2 school years before filter change needed	At least 2 school years before filter change needed	At least 2 school years before filter change needed
Electricity	9 watts on our recommended speed level	24 watts on our recommended speed level	22 watts on our recommended speed level	24 watts on our recommended speed level

Costs of clean air vs costs of not cleaning air

Where to learn more in McDaniel report: Page 136

K-8 schools

1st air purifier: Intellipure



2nd air purifier: Medify Pro



High schools

1st air purifier: Regular Medify

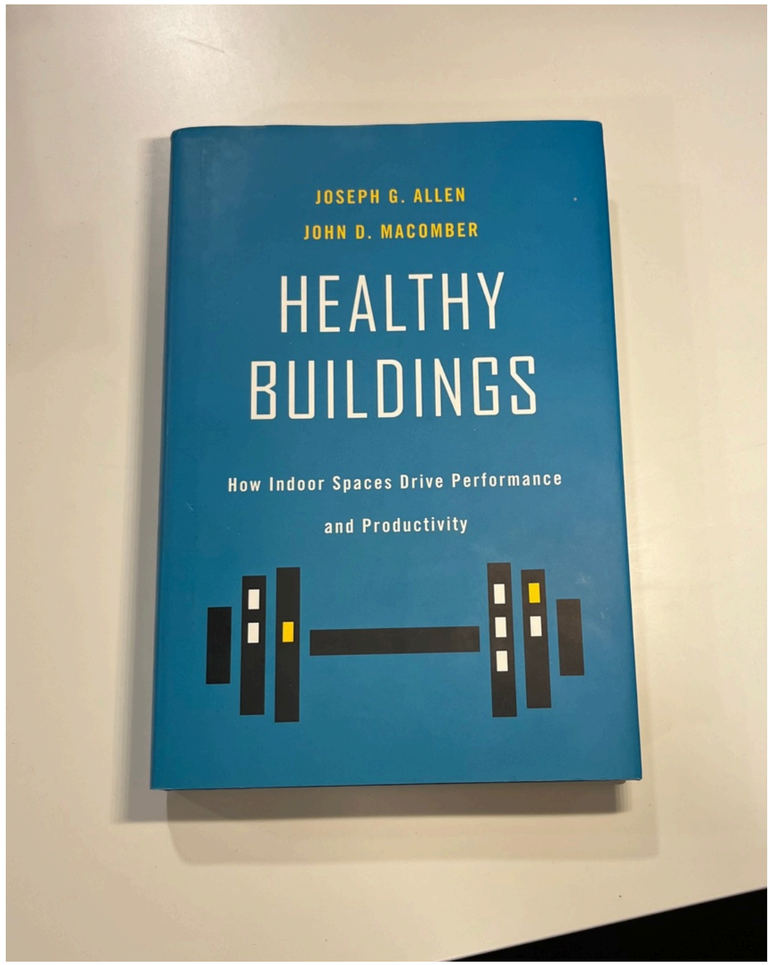


2nd air purifier: Medify Pro



	1st air purifier: Intellipure	2nd air purifier: Medify Pro	1st air purifier: Regular Medify	2nd air purifier: Medify Pro
Filter changes	6 school years before filter change needed	At least 2 school years before filter change needed	At least 2 school years before filter change needed	At least 2 school years before filter change needed
Electricity	9 watts on our recommended speed level	24 watts on our recommended speed level	22 watts on our recommended speed level	24 watts on our recommended speed level

Costs of clean air vs costs of not cleaning air



Drs. Joseph Allen & John Macomber writing in Healthy Buildings on

3-30-300 rule

For every:

- $3 spent on utilities like electricity and heat
- $30 is spent on rent or building ownership and construction
- $300 is spent on payroll

“This realization can make a focus on miserly utility spending, say, for ventilation, look pretty silly if the expensive assets – the humans – are not functioning at their best.”

Costs of clean air vs costs of not cleaning air

Monthly electricty cost to run two Medify air purifiers at our recommended levels in a PPS classroom

$1.17

Typical clean air added by two Medify air purifiers

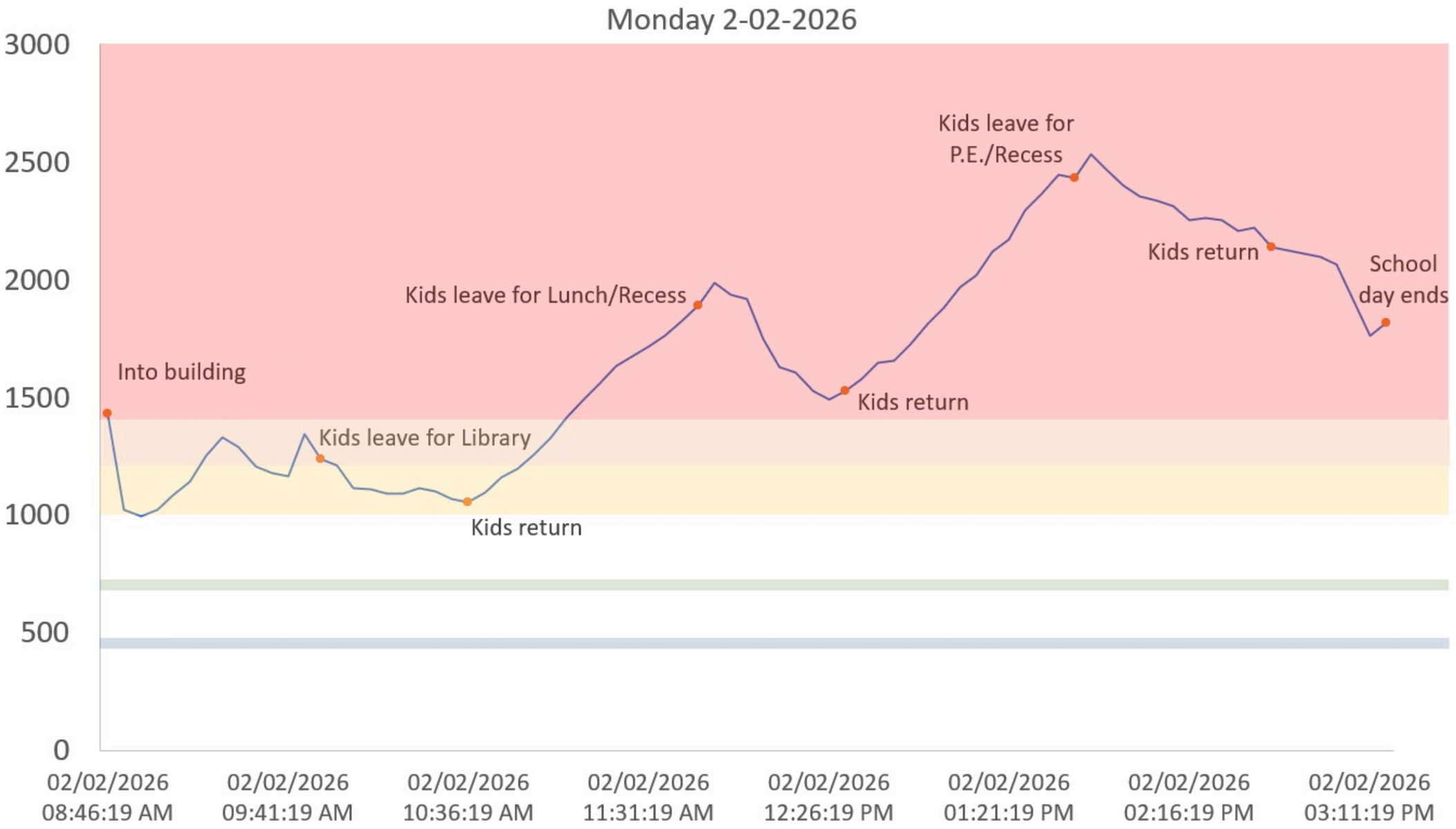
4

air changes/hour

Cost of a sub for 3 days

$827

New HVAC system at Bridger Creative Science

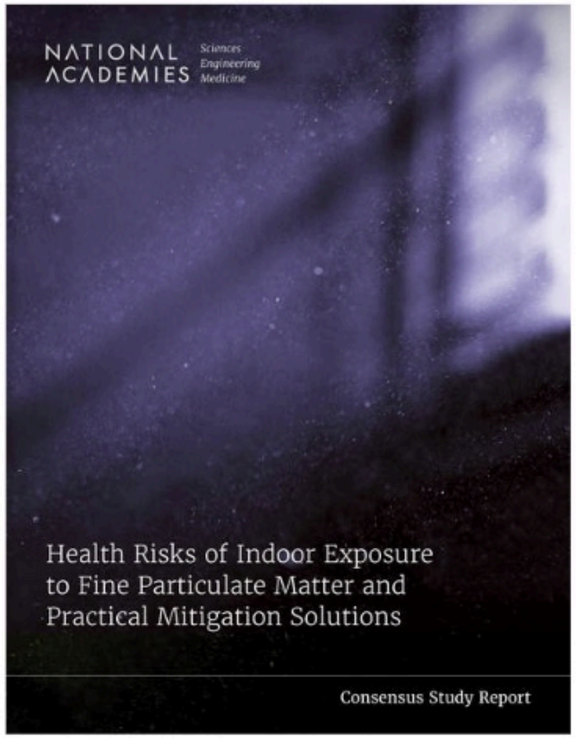


Dr. Richard Corsi, Ph.D., P.E.



Dean of the College of Engineering at the University of California, Davis

Chair of the National Academy of Sciences Committee on Health Risks of Indoor Exposure to Fine Particulate Matter and Practical Mitigation Solutions



270 publications & 3 decades of experience in IAQ research

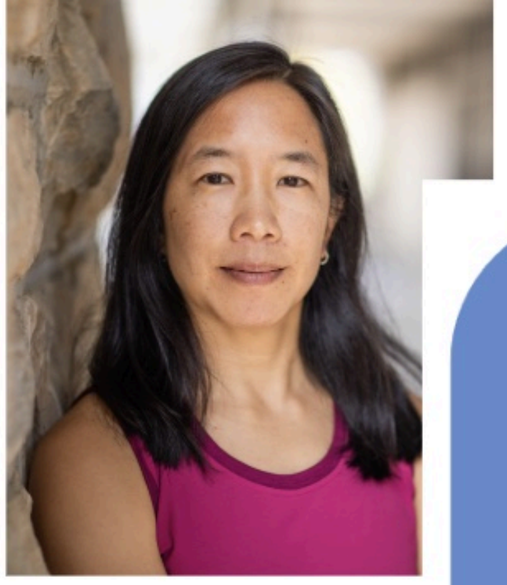
"I do not believe aiming for 4 or 6 ACH in schools or other indoor spaces is sufficiently aspirational & have written about how a combo of increased outdoor air supply and use of #CorsiRosenthalBox es can get us close to 10 equivalent ACH in classrooms."

Dr. Richard Corsi, Ph.D., PE.
Dean of Engineering at University of California, Davis

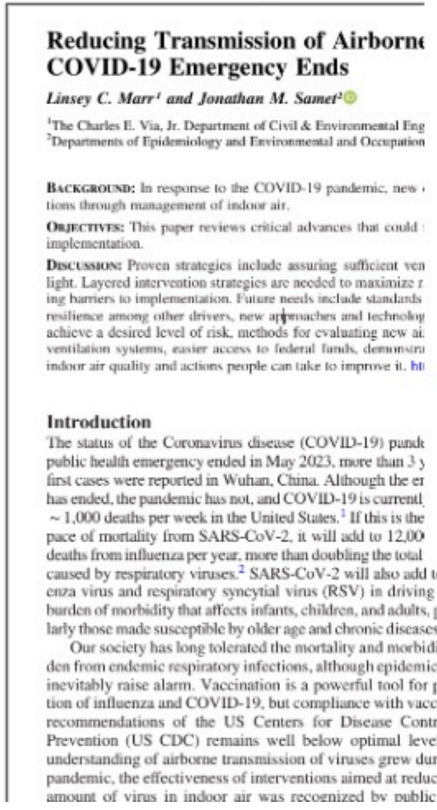
Dr. Linsey Marr, Ph.D., P.E.

University Distinguished Professor, Civil and Environmental Engineering, Virginia Tech University

MacArthur Fellow







Reducing Transmission of Airborne COVID-19 Emergency Ends

Linsey C. Marr[1] and Jonathan M. Samet[2]

[1]The Charles E. Via, Jr. Department of Civil & Environmental Eng
[2]Departments of Epidemiology and Environmental and Occupation

Background: In response to the COVID-19 pandemic, new
tions through management of indoor air.

Objectives: This paper reviews critical advances that could
implementation.

Discussion: Proven strategies include assuring sufficient ven
light. Layered intervention strategies are needed to maximize
ing barriers to implementation. Future needs include standards
resilience among other drivers, new approaches and technolog
achieve a desired level of risk, methods for evaluating new ai
ventilation systems, easier access to federal funds, demonstra
indoor air quality and actions people can take to improve it. hu

Introduction

The status of the Coronavirus disease (COVID-19) pande
public health emergency ended in May 2023, more than 3 y
first cases were reported in Wuhan, China. Although the er
has ended, the pandemic has not, and COVID-19 is currentl
~1,000 deaths per week in the United States.[1] If this is the
pace of mortality from SARS-CoV-2, it will add to 12,00
deaths from influenza per year, more than doubling the total
caused by respiratory viruses.[2] SARS-CoV-2 will also add to influenza virus and respiratory syncytial virus (RSV) in driving a huge burden of morbidity that affects infants, children, and adults, particularly those made susceptible by older age and chronic diseases.[3]

Our society has long tolerated the mortality and morbidity burden from endemic respiratory infections, although epidemic surges inevitably raise alarm. Vaccination is a powerful tool for prevention of influenza and COVID-19, but compliance with vaccination recommendations of the US Centers for Disease Control and Prevention (US CDC) remains well below optimal levels.[4] As understanding of airborne transmission of viruses grew during the pandemic, the effectiveness of interventions aimed at reducing the amount of virus in indoor air was recognized by public health

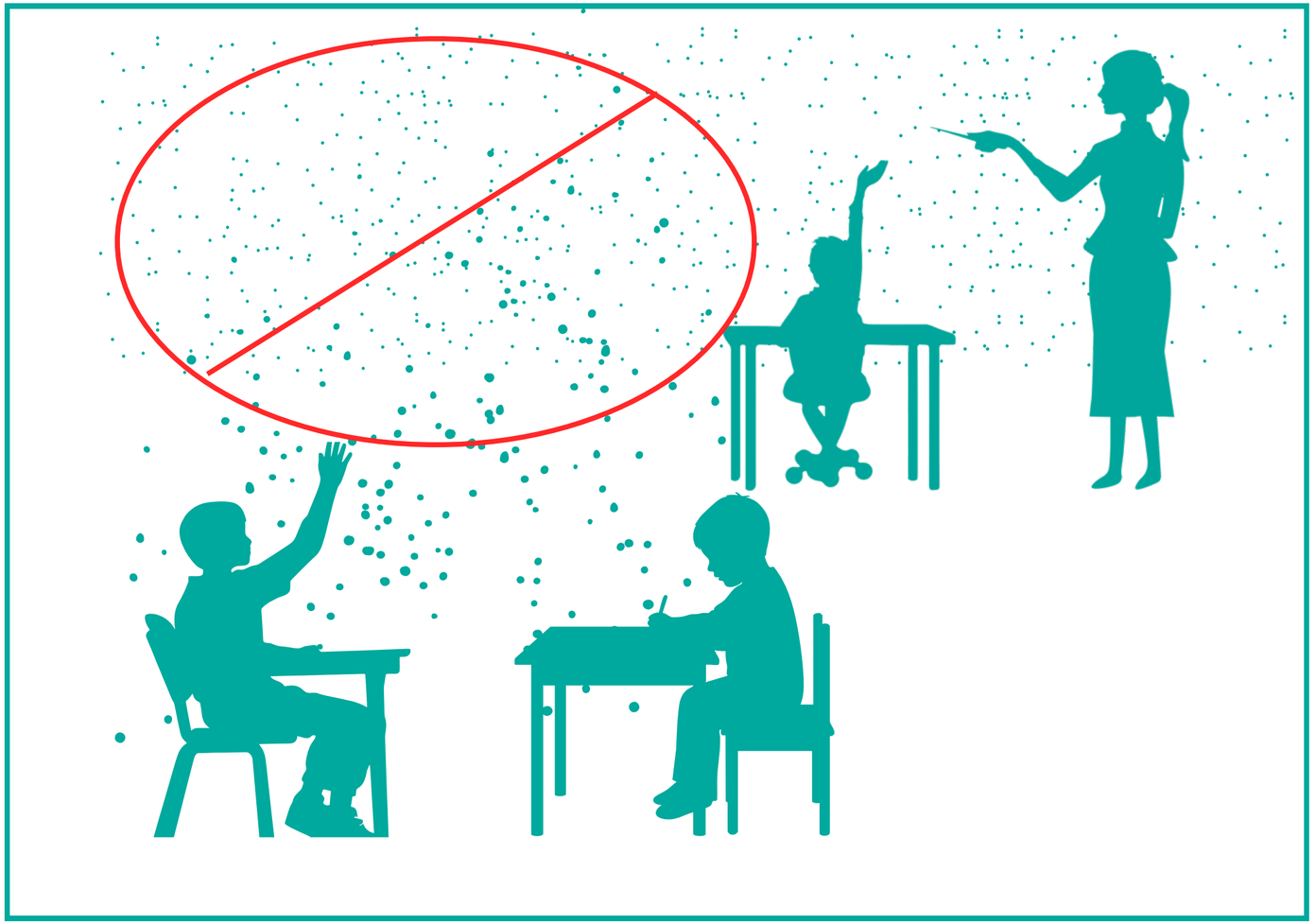
and transportation environments. The first workshop was held in August 2020,[5] when airborne transmission of SARS-CoV-2 was still being debated in spite of the strong evidence for it.[6,7] The second, a series of three sessions, was held in the fall of 2022, when research and field experience had accumulated to illuminate effective strategies for reducing airborne transmission of SARS-CoV-2 in indoor environments.[8] The second workshop included an initial session that focused on scientific advances, lessons learned, and organizational responses; the second session addressed schools; and the third session covered public transportation, excluding airplanes. We complement the workshop outcomes with selected reports on key findings. Since the last workshop, guidelines have advanced considerably in the form of the final report from the *Lancet* Commission on COVID-

"... schools [are] where I would prioritize resources ... I would go beyond five air changes per hour .... Let's hit the hospital grade ... 12 air changes per hour or better"

Dr. Linsey Marr, Ph.D., PE.
University Distinguished Professor, Civil and Environmental Engineering, Virginia Tech University
MacArthur Fellow

Designing for clean air: Best health analogies

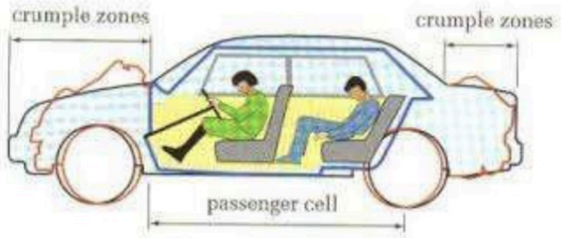
- We need to clean our indoor air the way we clean our water
- In drinking water treatment, the goal is to remove all traces of human waste
- The same commitment to filtration, purification, and safety for bodily fluids in indoor air should be our north star

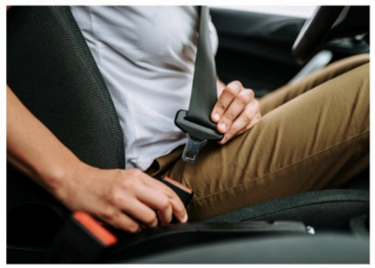


Where to learn more in McDaniel report: Page 10

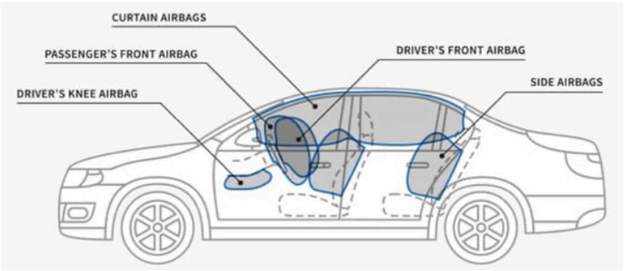
Designing for clean air: Best health analogies

- We also need to make indoor air safety work the way motor vehicle safety does
- In air quality, as in motor vehicle safety, no single feature is enough
- There are many car safety features we use together to prevent harm
- Examples:
  - Seat belts & car seats
  - Air bags
  - Safety glass windshields & windows
  - Dashboard/warning lights/blinkers
  - Crumple zones & side door protection









There are many car safety features we use together to prevent harm

We should also use many indoor air safety features to prevent harm

Examples

Seat belts & car seats →	Clean airflow Air changes per hour, CFM, etc
Air bags →	Carbon dioxide
Safety glass windshields & windows →	MERV 13+ furnace filters
Dashboard/warning lights →	In-room air quality monitors / sensors with displays to occupants & public
Crumple zones & side door protection →	In-room filtration
Interior passive safety design →	Occupancies used in design

Where to learn more in McDaniel report: Page 10

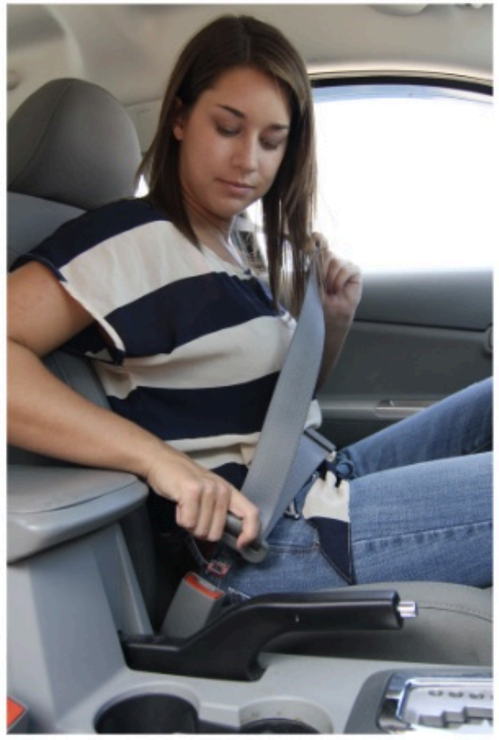
Minimally acceptable ventilation
(ASHRAE 62.1)



ASHRAE 62.1 + 30%



ASHRAE + 30% plus recirculation through MERV 13 to reach ~6 effective air changes per hour



12 ACHe where easy & energy-efficient = rear-facing to age 4

